3.5 - Curve Sketching

Goals: Let's put everything together and now graph functions based on all the information we know

Before Class Videos: No video for this section; consult Section 3.5 of APEX Calculus to review.

Example 1: Graph

$$f(x) = \frac{x}{x-5}, \qquad f'(x) = -\frac{5}{(x-5)^2}, \qquad f''(x) = \frac{10}{(x-5)^3}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

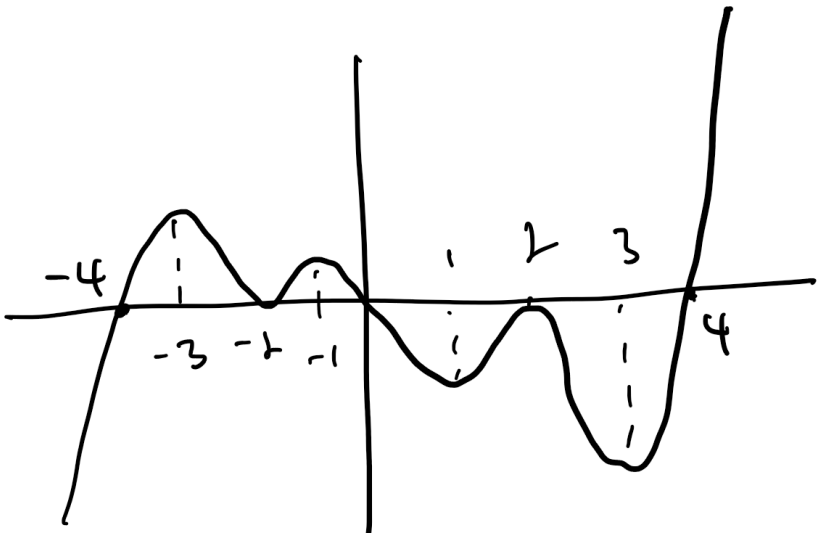
- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Let's graph it:

Example 2: Consider this plot of the derivative $f'(x)$ and classify all Critical Numbers



Example 3:

$$f(x) = \frac{1}{1+e^{-x}}, \qquad f'(x) = \frac{e^{-x}}{(1+e^{-x})^2}, \qquad f''(x) = \frac{e^{-x}(e^{-x}-1)}{(e^{-x}+1)^3}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Example 4: Graph

$$f = e^{1/x}, \qquad f'(x) = -\frac{e^{1/x}}{x^2}, \qquad f''(x) = \frac{e^{1/x}(2x+1)}{x^4}.$$

- Domain:

- Intercepts:

- Critical points:

- Vertical Asymptotes:

- Horizontal Asymptotes:

- Increasing/Decreasing:

- Concave Up/Down:

Let's plot it:

Example 5: (Time permitting) Graph

$$\frac{x-x^2}{x^2-9x+8}, \qquad f'(x)=\frac{x-4}{x^2-10x+25}, \qquad f''(x)=\frac{8}{(x-5)^3}$$

Suggested after class problems from APEX: Section 3.5: 11,15,21